

Appendix: Syntax by Program

Stata

```
* ----- *
* set working directory, import and view data
* ----- *
cd "/Users/Desktop/"
import delimited "cleaned_data.csv", clear
browse
* ----- *

* ----- *
* Model 0: Naive Approach Ignoring Changes in Group Membership (Pure 3-level Clustering)
* ----- *
* fit model
mixed math time female ses public suspension platform ///
|| school: , covariance(identity) ///
|| student: time, cov(unstructured) ///
reml variance
* ----- *

* ----- *
* Model 1: Acute Effects CCREM
* ----- *
* fit model
mixed math time female ses public suspension platform ///
|| _all: R.school, nocons covariance(identity) ///
|| student: time, cov(unstructured) ///
reml variance
* ----- *

* ----- *
* Model 2: Cumulative Effects CCREM
* ----- *
* to construct school Z matrix of cumulative effects, first create dummy
* indicators for each school
tabulate school, generate(d)

* for cumulative weights, fill downward consecutively
* (must run both commands [local] and [forvalues] simultaneously)
local n_schools = r(r)
forvalues x = 1/`n_schools' {
    gen w`x' = d`x'
    bysort student: replace w`x' = w`x'[_n-1] if w`x'==0 & w`x'[_n-1]==1
}

* fit model
mixed math time female ses public suspension platform ///
```

```

|| _all: w*, nocons covariance(identity) ///
|| student: time, cov(unstructured) ///
reml variance
* ----- *

* ----- *
* Model 3: Cumulative Effects CCREM with Standardized Weights
* ----- *
* keep same weights from previous Model 2, but ensure weights are proportional
* such that weights sum to 1 for each row of the weight matrix
egen tot_w = rowtotal(w*)
foreach x of varlist w* {
    replace `x' = `x' / tot_w
}

* fit model
mixed math time female ses public suspension platform ///
|| _all: w*, nocons covariance(identity) ///
|| student: time, cov(unstructured) ///
reml variance
* ----- *

* ----- *
* Model 4: Acute Effects CCREM with Autoregressive Group Effects
* ----- *

* not able to implement

* ----- *

* ----- *
* Model 5: Acute Effects CCREM with Compound Symmetric Group Effects
* ----- *

* not able to implement

* ----- *

* ----- *
* Model 6: Acute Effects CCREM with Unstructured Group Effects
* ----- *

* not able to implement

* ----- *

```

SAS

```
/* ----- */
/* imoprt and view data
/* ----- */
proc import datafile="/home/cleaned_data.csv"
dbms = csv
out = data_long
replace;
proc print data=data_long(obs=10);
/* ----- */

/* ----- */
/* Model 0: Naive Approach Ignoing Changes in Group Membership (Pure 3-level Clustering)
/* ----- */
/* fit model */
proc mixed data=data_long method=reml;
class school student;
model math = time female ses public suspension platform / solution;
random intercept / subject = school; *g gcorr;
random intercept time / subject = student(school) type=un;
run;
/* ----- */

/* ----- */
/* Model 1: Acute Effects CCREM
/* ----- */
/* fit model */
proc mixed data=data_long method=reml;
class school student;
model math = time female ses public suspension platform / solution;
random intercept / subject=school; *g gcorr;
random intercept time / subject=student type=un gcorr;
run;
/* ----- */

/* ----- */
/* Model 2: Cumulative Effects CCREM
/* ----- */
/* first, define function to construct Z matrix of cumulative effects */
/* (Cafri et al., 2015) */
%macro cummulativematrixformat(datain=, personID=, grpname=, outcome=, dataout=);

proc freq data=&datain;
tables &grpname/out=w;
run;
```

```

ods output summary=group;
proc means data=w n;
var &grpname;
run;

data group2;
set group;
call symput("groupN", trim(left(put(&grpname._N,best.))));
run;
%put groupN =====> |&groupN| ;

proc glimmix data=&datain outdesign(z)=zmat;
class &grpname;
model &outcome = ;
random &grpname ;
run;

proc sort data=zmat;
by &personID;
run;

data acute ;
set zmat ;
array glimout _z1-_z&groupN;
array recode x1-x&groupN;
by &personID;
do i=1 to &groupN;
    if glimout{i} > 0 then do; recode{i}=glimout{i}; recode{i}=i; end;
    if glimout{i} = 0 then do; recode{i}=glimout{i}; end;
end;
drop i _z1-_z&groupN;
run;

data cumulative1 ;
set acute ;
array origvar x1-x&groupN;
array carry d1-d&groupN;
by &personID;
retain carry;
do i=1 to &groupN;
    if first.&personID then carry{i} = . ;
end;
do i=1 to &groupN;
    if origvar{i} > 0 then do;carry{i}=origvar{i};end;
end;
run;

data cumulative2;
set cumulative1;
array origvar x1-x&groupN;
array carry d1-d&groupN;
do i=1 to &groupN;
    if origvar{i} = . then carry{i} = . ;

```

```

        if origvar{i} = 0 and carry{i} = . then carry{i}=0 ;
    end;
run;

data &dataout ;
set cumulative2;
array carry d1-d&groupN;
array weight w1-w&groupN;
do i=1 to &groupN;
    if carry{i}>0 then weight{i}=1;
    else weight{i}=carry{i};
end;
drop i;
run;
%mend;

/* apply to data, observe new data */
%cummulativeformat(
datain = data_long,
personID = student,
grpname = school,
outcome = math,
dataout = data_long2
);
proc print data=data_long2(obs=10);

/* fit model */
proc glimmix data=data_long2;
class d1-d83 student;
effect school2 = multimeter(d1-d83 / weight=(w1-w83) details);
model math = time female ses public suspension platform / dist=gaussian link=identity solution;
random school2; *g gcorr;
random intercept time / subject=student type=un;
run;
/* ----- */

/* ----- */
/* Model 3: Cumulative Effects CCREM with Standardized Weights
/* ----- */
/* fit model */
proc glimmix data=data_long2;
class d1-d83 student;
effect school2 = multimeter(d1-d83 / weight=(w1-w83) stdsize details);
model math = time female ses public suspension platform / dist=gaussian link=identity solution;
random school2; *g gcorr;
random intercept time / subject=student type=un;
run;
/* ----- */

```

```

/* ----- */
/* Model 4: Acute Effects CCREM with Autoregressive Group Effects
/* ----- */
/* must first make duplicate time variable named time2 */
data data_long;
set data_long;
time2 = time;
run;
proc print data=data_long(obs=10);

/* fit model */
proc mixed data=data_long method=reml;
class school student time2;
model math = time female ses public suspension platform / solution;
random time2 / subject=school type=ar(1); *g gcorr;
random intercept time / subject=student type=un;
run;
/* ----- */

/* ----- */
/* Model 5: Acute Effects CCREM with Compound Symmetric Group Effects
/* ----- */
/* fit model */
proc mixed data=data_long method=reml;
class school student time2;
model math = time female ses public suspension platform / solution;
random time2 / subject=school type=cs; *g gcorr;
random intercept time / subject=student type=un;
run;
/* ----- */

/* ----- */
/* Model 6: Acute Effects CCREM with Unstructured Group Effects
/* ----- */
/* fit model */
proc mixed data=data_long method=reml;
class school student time2;
model math = time female ses public suspension platform / solution;
random time2 / subject=school type=un; *g gcorr;
random intercept time / subject=student type=un;
run;
/* ----- */

```

lme4

```
# ----- #
# load necessary packages, set working directory, import and view data
# ----- #
rm(list = ls()); gc()
library("lme4")
library("lmerTest")

options(scipen = 99)
set.seed(5543)

setwd("/Users/Desktop")
dlong <- read.csv("cleaned_data.csv")
head(dlong, 10)
# ----- #

# ----- #
# Model 0: Naive Approach Ignoring Changes in Group Membership (Pure 3-level Clustering)
# ----- #
# fit model
m0 <- lmer(math ~ time + female + ses + public + suspension + platform +
            (1 | school) +
            (1 + time | school:student),
            data = dlong, REML = TRUE)

# view parameter estimates
summary(m0)

# view random effects
print(VarCorr(m0), comp = "Variance")

# covariance matrix of student random effects
as.matrix(bdiag(VarCorr(m0)[1]))

# covariance matrix of school random effects
as.matrix(bdiag(VarCorr(m0)[2]))
# ----- #

# ----- #
# Model 1: Acute Effects CCREM
# ----- #
# fit model
m1 <- lmer(math ~ time + female + ses + public + suspension + platform +
            (1 | school) +
            (1 + time | student),
            data = dlong, REML = TRUE)

# view parameter estimates
```



```

summary(m1)

# view random effects
print(VarCorr(m1), comp = "Variance")

# covariance matrix of student random effects
as.matrix(bdiag(VarCorr(m1)[1]))

# covariance matrix of school random effects
as.matrix(bdiag(VarCorr(m1)[2]))
# ----- #

# ----- #
# Model 2: Cumulative Effects CCREM
# ----- #

# not able to implement
# ----- #

# ----- #
# Model 3: Cumulative Effects CCREM with Standardized Weights
# ----- #

# not able to implement
# ----- #

# ----- #
# Model 4: Acute Effects CCREM with Autoregressive Group Effects
# ----- #

# not able to implement
# ----- #

# ----- #
# Model 5: Acute Effects CCREM with Compound Symmetric Group Effects
# ----- #

# not able to implement
# ----- #

```

```

# ----- #
# Model 6: Acute Effects CCREM with Unstructured Group Effects
# ----- #
# fit model
m6 <- lmer(math ~ time + female + ses + public + suspension + platform +
           (0 + as.factor(time) | school) +
           (1 + time | student),
           data = dlong, REML = TRUE)

# view parameter estimates
summary(m6)

# view random effects
print(VarCorr(m6), comp = "Variance")

# covariance matrix of student random effects
as.matrix(bdiag(VarCorr(m6)[1]))

# covariance matrix of school random effects
as.matrix(bdiag(VarCorr(m6)[2]))
# ----- #

```

nlme

```
# ----- #
# load necessary packages, set working directory, import and view data
# ----- #
rm(list = ls()); gc()
library("nlme")
library("lmerTest")
library("Matrix")
library("tidyverse")
library("fastDummies")

options(scipen = 99)
set.seed(5543)

# define function to extract random effects covariance matrices
get_re_covariance_matrix <- function(model) {

  # extract SDs
  school_sd <- unique(attr(corMatrix(model$modelStruct[[1]])[[1]], which = "stdDev"))
  student_sd <- unique(attr(corMatrix(model$modelStruct[[1]])[[2]], which = "stdDev"))

  # multiply by residual
  school_sd <- school_sd * model$sigma
  student_sd <- student_sd * model$sigma

  # square to calculate variance
  school_var <- school_sd^2
  student_var <- student_sd^2

  # calculate covariance b/w intercept and slope for student
  student_correlation <- corMatrix(model$modelStruct[[1]])[[2]][2,1]
  student_covar <- (student_sd[1] * student_sd[2]) * student_correlation

  school_matrix <- matrix(school_var)
  student_matrix <- matrix(c(student_var[1],
                             student_covar,
                             student_covar,
                             student_var[2]),
                          nrow = 2, ncol = 2, byrow = FALSE)

  return(list("school_covariance_matrix" = school_matrix,
             "student_covariance_matrix" = student_matrix,
             "residual_variance" = (model$sigma)^2))
}

setwd("/Users/Desktop")
dlong <- read.csv("cleaned_data.csv")
head(dlong, 10)
# ----- #
```

```

# ----- #
# Model 0: Naive Approach Ignoring Changes in Group Membership (Pure 3-level Clustering)
# ----- #
# fit model
m0 <- lme(math ~ time + female + ses + public + suspension + platform,
  random = list(
    school = ~ 1,
    student = ~ 1 + time
  ),
  data = dlong, method = "REML")

# view parameter estimates
summary(m0)

# view random effects
print(VarCorr(m0), comp = "Variance")
get_re_covariance_matrix(m0)
# ----- #

# ----- #
# Model 1: Acute Effects CCREM
# ----- #
# first define dummy (factor) variable == 1
dlong$DummyOf1s <- factor(1)

# fit model
m1 <- lme(math ~ time + female + ses + public + suspension + platform,
  random = list(
    DummyOf1s = pdIdent(~ 0 + factor(school)),
    student = pdSymm(~ 1 + time)
  ),
  data = dlong, method = "REML")

# view parameter estimates
summary(m1)

# view random effects
print(VarCorr(m1), comp = "Variance")
get_re_covariance_matrix(m1)
# ----- #

# ----- #
# Model 2: Cumulative Effects CCREM
# ----- #
# to construct school Z matrix of cumulative effects, first create dummy
# indicators for each school
dummy <- fastDummies::dummy_cols(dlong$school)

```

```

dummy <- dummy[,-1]
colnames(dummy) <- paste0("d", 1:length(unique(dlong$school)))

# view dummy matrix
head(dummy)

# make weight indicators for each school
weight <- ifelse(dummy == 1, 1, NA)
colnames(weight) <- paste0("w", 1:length(unique(dlong$school)))

# view weight matrix
head(weight)

# add dummy and weight indicators as new columns to original data
dlong <- cbind(dlong, dummy, weight)

# for cumulative weights, fill downward consecutively
dlong <- dlong %>%
  dplyr::group_by(student) %>%
  tidyr::fill(w1:paste0("w", length(unique(dlong$school))), .direction="down") %>%
  ungroup()
dlong[,grep("^w", colnames(dlong))] <- ifelse(is.na(dlong[,grep("^w", colnames(dlong))]) == TRUE, 0, 1)
dlong <- data.frame(dlong)

# remove unnecessary items, and view data with new variables
rm(dummy,weight)
head(dlong)

# the following is the formula used for group random effects
ranef_formula <- paste0(paste0("w",1:length(unique(dlong$school))), collapse = "+")
reformulate(ranef_formula)

# need to define dummy (factor) variable == 1
dlong$DummyOf1s <- factor(1)

# fit model
m2 <- lme(math ~ time + female + ses + public + suspension + platform,
  random = list(
    DummyOf1s = pdIdent(reformulate(ranef_formula)),
    student = pdSymm(~ 1 + time)
  ),
  data = dlong, method = "REML")

# view parameter estimates
summary(m2)

# view random effects
get_re_covariance_matrix(m2)
# ----- #

# ----- #

```

```

# Model 3: Cumulative Effects CCREM with Standardized Weights
# ----- #
# keep same weights from previous Model 2, but ensuring weights are proportional
# such that weights sum to 1 for each row of the weight matrix

# first, count the number of weights == 1 in a given row
dlong$tot_w <- rowSums(dlong[,grep("^w", colnames(dlong))])

# then, each weight gets divided by the row total
dlong[,grep("^w",colnames(dlong))] <- lapply(
  dlong[,grep("^w",colnames(dlong))],
  function(x) x / dlong[,"tot_w"]
)

# view data
tail(dlong, 10)

# need to define dummy (factor) variable == 1
dlong$DummyOfis <- factor(1)

# fit model
m3 <- lme(math ~ time + female + ses + public + suspension + platform,
  random = list(
    DummyOfis = pdIdent(reformulate(ranef_formula)),
    student = pdSymm(~ 1 + time)
  ),
  data = dlong, method = "REML")

# view parameter estimates
summary(m3)

# view random effects
get_re_covariance_matrix(m3)
# ----- #

# ----- #
# Model 4: Acute Effects CCREM with Autoregressive Group Effects
# ----- #

# not able to implement
# ----- #

# ----- #
# Model 5: Acute Effects CCREM with Compound Symmetric Group Effects
# ----- #

# not able to implement

```

```
# ----- #  
  
# ----- #  
# Model 6: Acute Effects CCREM with Unstructured Group Effects  
# ----- #  
  
# not able to implement  
  
# ----- #
```

glmmTMB

```
# ----- #
# load necessary packages, set working directory, import and view data
# ----- #
rm(list = ls()); gc()
library("glmmTMB")
library("lmerTest")
library("Matrix")
library("tidyverse")
library("fastDummies")

options(scipen = 99)
set.seed(5543)

setwd("/Users/Desktop")
dlong <- read.csv("cleaned_data.csv")
head(dlong, 10)
# ----- #

# ----- #
# Model 0: Naive Approach Ignoring Changes in Group Membership (Pure 3-level Clustering)
# ----- #
# fit model
m0 <- glmmTMB(math ~ time + female + ses + public + suspension + platform +
              (1 | school) +
              (1 + time | school:student),
              data = dlong, REML = TRUE)

# view parameter estimates
summary(m0)

# view random effects
print(VarCorr(m0), comp = "Variance")
VarCorr(m0)$cond$school
VarCorr(m0)$cond$`school:student`
# ----- #

# ----- #
# Model 1: Acute Effects CCREM
# ----- #
# fit model
m1 <- glmmTMB(math ~ time + female + ses + public + suspension + platform +
              (1 | school) +
              (1 + time | student),
              data = dlong, REML = TRUE)

# view parameter estimates
summary(m1)
```



```

# view random effects
print(VarCorr(m1), comp = "Variance")
VarCorr(m1)$cond$school
VarCorr(m1)$cond$student
# ----- #

# ----- #
# Model 2: Cumulative Effects CCREM
# ----- #
# to construct school Z matrix of cumulative effects, first create dummy
# indicators for each school
dummy <- fastDummies::dummy_cols(dlong$school)
dummy <- dummy[,-1]
colnames(dummy) <- paste0("d", 1:length(unique(dlong$school)))

# view dummy matrix
head(dummy)

# make weight indicators for each school
weight <- ifelse(dummy == 1, 1, NA)
colnames(weight) <- paste0("w", 1:length(unique(dlong$school)))

# view weight matrix
head(weight)

# add dummy and weight indicators as new columns to original data
dlong <- cbind(dlong, dummy, weight)

# for cumulative weights, fill downward consecutively
dlong <- dlong %>%
  dplyr::group_by(student) %>%
  tidyr::fill(w1:paste0("w", length(unique(dlong$school))), .direction="down") %>%
  ungroup()
dlong[,grep("^w", colnames(dlong))] <- ifelse(is.na(dlong[,grep("^w", colnames(dlong))]) == TRUE, 0, 1)
dlong <- data.frame(dlong)

# remove unnecessary items, and view data with new variables
rm(dummy,weight)
head(dlong)

# the following is the formula used for group random effects
ranef_formula <- paste0(paste0("w",1:length(unique(dlong$school))), collapse = "+")
ranef_formula

# need to define dummy (factor) variable == 1
dlong$DummyOf1s <- factor(1)

# fit model
m2 <- glmmTMB(as.formula(paste0("math ~ time + female + ses + public + suspension + platform +
                                homdiag(",ranef_formula," | DummyOf1s) +
                                (1 + time | student)")),

```

```

        data = dlong, REML = TRUE)

# view parameter estimates
summary(m2)

# view random effects
print(VarCorr(m2), comp="Variance")
VarCorr(m2)$cond$DummyOf1s
VarCorr(m2)$cond$student
# ----- #

# ----- #
# Model 3: Cumulative Effects CCREM with Standardized Weights
# ----- #
# keep same weights from previous Model 2, but ensuring weights are proportional
# such that weights sum to 1 for each row of the weight matrix

# first, count the number of weights == 1 in a given row
dlong$tot_w <- rowSums(dlong[,grep("^w", colnames(dlong))])

# then, each weight gets divided by the row total
dlong[,grep("^w", colnames(dlong))] <- lapply(
  dlong[,grep("^w", colnames(dlong))],
  function(x) x / dlong[, "tot_w"]
)

# view data
tail(dlong, 10)

# need to define dummy (factor) variable == 1
dlong$DummyOf1s <- factor(1)

# fit model
m3 <- glmmTMB(as.formula(paste0("math ~ time + female + ses + public + suspension + platform +
                                homdiag(", raneff_formula, " | DummyOf1s) +
                                (1 + time | student)")),
              data = dlong, REML = TRUE)

# view parameter estimates
summary(m3)

# view random effects
print(VarCorr(m3), comp="Variance")
VarCorr(m3)$cond$DummyOf1s
VarCorr(m3)$cond$student
# ----- #

# ----- #
# Model 4: Acute Effects CCREM with Autoregressive Group Effects

```

```

# ----- #
# fit model
m4 <- glmmTMB(math ~ time + female + ses + public + suspension + platform +
              ari(0 + factor(time) | school) +
              (1 + time | student),
              data = dlong, REML = TRUE)

# view parameter estimates
summary(m4)

# view random effects
print(VarCorr(m4), comp="Variance")
VarCorr(m4)$cond$school
VarCorr(m4)$cond$student
# ----- #

# ----- #
# Model 5: Acute Effects CCREM with Compound Symmetric Group Effects
# ----- #
# fit model
m5 <- glmmTMB(math ~ time + female + ses + public + suspension + platform +
              cs(0 + factor(time) | school) +
              (1 + time | student),
              data = dlong, REML = TRUE,
              map = list(theta = factor(c(1,1,1,1,4,5,6,7))))

# view parameter estimates
summary(m5)

# view random effects
print(VarCorr(m5), comp="Variance")
VarCorr(m5)$cond$school
VarCorr(m5)$cond$student
# ----- #

# ----- #
# Model 6: Acute Effects CCREM with Unstructured Group Effects
# ----- #
# fit model
m6 <- glmmTMB(math ~ time + female + ses + public + suspension + platform +
              (0 + as.factor(time) | school) +
              (1 + time | student),
              data = dlong, REML = TRUE)

# view parameter estimates
summary(m6)

# view random effects
print(VarCorr(m6), comp="Variance")

```

```
VarCorr(m6)$cond$school
VarCorr(m6)$cond$student
# ----- #
```